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TdP

Python data structures and collections

A wide choice of containers for your data

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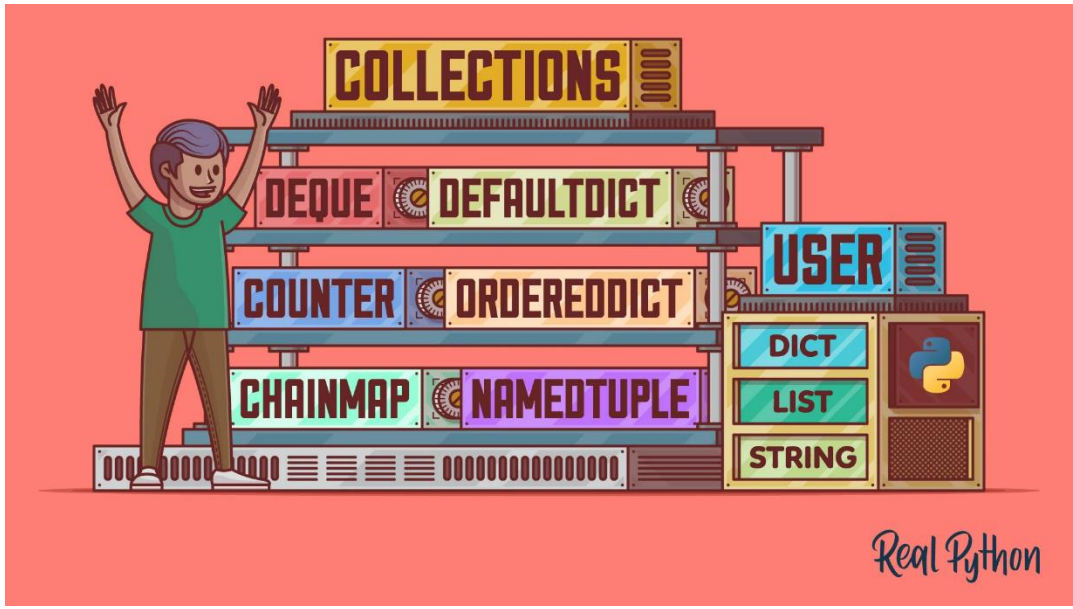

https://realpython.com/python-collections-module/
https://realpython.com/python-data-structures/
https://docs.python.org/3/library/collections.html
https://docs.python.org/3/library/datatypes.html

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Data structures

- The Python language offers very powerful built-in data structures
  - `list` and `tuple`
  - `set`
  - `dict`
- They can be used to store and search information, and each is specialized to support some use cases
- Additional data structures are available in the standard library, to cover other use cases

Overview

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Remember…

Schema sinottico delle principali operazioni sui contenitori					
Operation	str	list	tuple	set	dict
Create	"abc" 'abc'	[a, b, c]	(a, b, c)	{a, b, c}	{a:x, b:y, c:z}
Create empty	"" ''	[] list()	() tuple()	set()	{} dict()
Access i-th item	s[i]	l[i]	u[i]		d[k] d.get(k,default)
Modify i-th item		l[i]=x			d[k]=x
Add one item (modify value)		l.append(x)		t.add(x)	d[k]=x
Add one item at position (modify value)		l.insert(i,x)			
Add one item (return new value)	s+'x'	l+[x]	u+(x,)		
Join two containers (modify value)		l.extend(l1)		t.update(t1)	
Join two containers (return new value)	s+s1	l+l1	u+u1	t.union(t1) t|t1	
Does it contain a value?	x in s	x in l	x in u	x in s	k in d (search keys) x in d.values() (search values)
Where is a value? (returns index)	s.find(x) s.index(x)	l.index(x)	u.index(x)		
Delete an item, by index		l.pop(i) l.pop()			d.pop(k)
Delete an item, by value		l.remove(x)		t.remove(x) t.discard(x)	
Sort (modify value)		l.sort()			
Sort (return new **list**)	sorted(s)	sorted(l)	sorted(u)	sorted(t)	sorted(d) (keys) sorted(d.items())

https://polito-informatica.github.io/Materiale/CheatSheet/Python_Cheat_Sheet-3.2.pdf

Comparison and ordering

- Objects can be compared if they define an `__eq__` method
  - Used internally by == and `!=` operators
  - Used internally by `find`, `index`, `in`, …
- Objects can be ordered if they define a `__lt__` method (and optionally, other comparison dunder methods)
  - Must define __eq__, in addition
  - Used internally by `<` `<=` `>` `>=` operators
  - Used internally by `sort`, `sorted`

Special case: predefined types

- Some built-in collections already define __eq__ and __lt__, therefore they are comparable and sortable
  - str, list, tuple compare their elements in left-to-right order
  - The contained values must be comparable/sortable, too
- Dictionaries support __eq__ but not __lt__
  - dict object cannot be ordered
- Sets support __eq__, but define __lt__ to mean “subset”
  - Misleading, do not try to order a list of sets

Special case: dataclasses

- By default, a dataclass defines the `__eq__` method
  - To prevent, define it as `@dataclass(eq=False)`
- By default, a dataclass does not define the `__lt__` method
  - To generate it, define with `@dataclass(order=True)`: will generate `__lt__()`, `__le__()`, `__gt__()`, and `__ge__()`
  - Automaticaly generated methods compare all the fields of the object, in the order in which they are declared
  - One or more fields may be omitted from comparison and ordering methods, by inizializing them with `field(compare=False)`

Example

```python
@dataclass(order=True)
class Voto:
    esame: str
    cfu: int
    punteggio: int
    lode: bool
    data: str = field(compare=False)
```

Sorting by other criteria

- If you want to sort a collection using criteria different from the `__lt__` method (or if `__lt__` is not defined), use the `key=` argument
- `key=operator.itemgetter('keyname')`
  - sort by dictionary[‘keyname’]
- `key=operator.itemgetter(itemnumber)`
  - Sort by list/tuple[itemnumber]
- `key=operator.attrgetter('attrname')`
  - Sort by object.attrname
- `key = lambda obj: something(obj)`
  - Sort by value of something() function
  - Example: `lambda v: v.voto` is equivalent to `operator.attrgetter('voto')`

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Dictionaries

- Map a “key” to a “value”
  - Key: unique value of a hashable type
  - Value: any object
- `dict`
  - Very efficient, constant time for insertion, search, deletion
  - Retains insertion order of elements
  - Has built-in syntax `{ key: val }` for creation

`d[key] = value`	Set a new value for a key
`d[key]`	Retrieve value from the key. May raise `KeyError`
`d.clear()`	Clears a dictionary.
`d.get(key, default)`	Returns the value for a key if it exists in the dictionary. Otherwise, returns a default value
`d.items()`	Returns a list of key-value pairs in a dictionary.
`d.keys()`	Returns a list of keys in a dictionary.
`d.values()`	Returns a list of values in a dictionary.
`d.pop(key, default)`	Removes a key from a dictionary, if it is present, and returns its value. Otherwise, returns a default value
`d.popitem()`	Removes the last key-value pair from a dictionary.
`d.update(obj)`	Merges a dictionary with another dictionary

“Hashable”?

- A hashable object
  - Has a hash value that never changes during its lifetime (defines `__hash__`)
  - It can be compared to other objects (defines `__eq__`)
- Hashable objects that compare as equal <u>must</u> have the same hash value
  - $a == b \Rightarrow hash(a) == hash(b)$

- Note: instances of user-defined classes are hashable by default. They all compare unequal (except with themselves), and their hash value is derived from their id(). You can redefine this behavior

Hash functions

- A hash function is a function that maps any object into an integer number (over 64 bit)
- It is needed to quickly discover if two objects are
  - Surely different
  - Very likely equal
- Used in the hash() function and internally in set, frozenset and dict.

https://docs.python.org/3/reference/datamodel.html#object.__hash__

Other dictionaries

- `collections.defaultdict`
  - A class that automatically provides a default value for non-existent keys
  - Requires a “factory” function to build the default values: list, str, int, … or custom
  - `d = collections.defaultdict(int)`
- `types.MappingProxyType`
  - Creates a “read-only” dictionary, without copying it
  - `readonly_d = types.MappingProxyType(normal_d)`
  - All modifications will generate an exception
    - `TypeError: 'mappingproxy' object does not support item assignment`

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Main Array types

- `list`
  - The most versatile one, mutable ordered sequence of objects of any value
  - Indexed by number (0…len()-1)
- `tuple`
  - An immutable version of a list: elements cannot be added, removed nor replaced
    - But… elements can be mutated, if they are mutable
  - Hashable, if its elements are hashable
- `str`
  - An array of Unicode Characters
  - Immutable

Specialized Array types

- `array.array`
  - Implemented in C as an array of elements of the same basic type (byte, int, float)
  - The type is declared at the time of creation
    - `arr = array.array("f", (1.0, 1.5, 2.0, 2.5))`
  - Uses less memory than normal lists, but less versatile
- `bytes`: Immutable Arrays of Single Bytes
- `bytearray`: Mutable Arrays of Single Bytes

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Records

- A record is a collection of data of different types, and different meanings, grouped together to represent a single high-level information

```
car = {
  "type": "Panda",
  "year": 2010
}
```

Implemented as…

dict

class

```
class Car:
    def __init__(self, type, year):
        self.type = type
        self.year = year
```

tuple

```
car = ("Panda", 2010)
```

dataclass

```
@dataclass
class Car:
    type: str
    year: int
```

Specialized record types

- `collections.namedtuple`
  - A tuple whose indices are not integers, but attributes (like objects)

```
Car = collections.namedtuple("Car", ("name", "year"))
c1 = Car("Panda", 2010)
c1.name  # ‘Panda’
```

  - Attribute values are immutable
- `typing.NamedTuple`
  - Uses a syntax similar to dataclasses

```
class Car(typing.NamedTuple):
    name: str
    year: int
```

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Sets

- `set`
  - Mutable container of hashable objects.
  - Duplicates are not allowed.
  - Simple syntax: `{ 1, 2, 3 }`
  - Supports set-theory operations
- `frozenset`
  - An immutable version of a set: once created, its elements cannot be changed
  - Since it’s hashable, it may be used as a key in a dictionary (or as an element in a set)

Multisets and `collections.Counter`

- The Counter class is useful for computing and storing frequencies of items (i.e. counts of elements that may appear more than once in a set)

```
cnt = collections.Counter([1, 2, 3, 3, 4, 5, 1, 8, 3, 5, 2,
2, 3, 8])
Counter({3: 4, 2: 3, 1: 2, 5: 2, 8: 2, 4: 1})
```

- Great for statistics, frequency counting, histogram, duplicate detection, ranking, …
- Internally stored as a defaultdict, with keys at the set elements, and values as the occurrence counts, with default value = 0

https://docs.python.org/3/library/collections.html#counter-objects

Creating `Counter` objects

- `c = Counter()` `# a new, empty counter`
- `c = Counter('gallahad')` `# a new counter from an iterable`
- `c = Counter(['eggs', 'ham'])` `# a new counter from an iterable`
- `c = Counter({'red': 4, 'blue': 2})` `# a new counter from a mapping`
- `c = Counter(cats=4, dogs=8)` `# a new counter from keyword args`

- Manually increasing counts:

```
for word in ['red', 'blue', 'red', 'green', 'blue', 'blue']:
    cnt[word] += 1
```

equivalent to

```
cnt = Counter(['red', 'blue', 'red', 'green', 'blue', 'blue'])
```

What can I do with a `Counter`?

- `c.most_common(n)` # the ‘n’ (default: all) most common items
- `c.total()` # total of all counts
- `list(c)` # list unique elements

- `set(c)` # convert to a set
- `dict(c)` # convert to a regular dictionary
- `c.items()` # convert to a list of (elem, cnt) pairs
- `c.elements()` # return a list [elem, …] with repetitions
- `Counter(dict(list_of_pairs))` # convert from a list of (elem, cnt) pairs
- `c.most_common()[:-n-1:-1]` # n least common elements
- `+c` # remove zero and negative counts
- `c.clear()` # reset all counts

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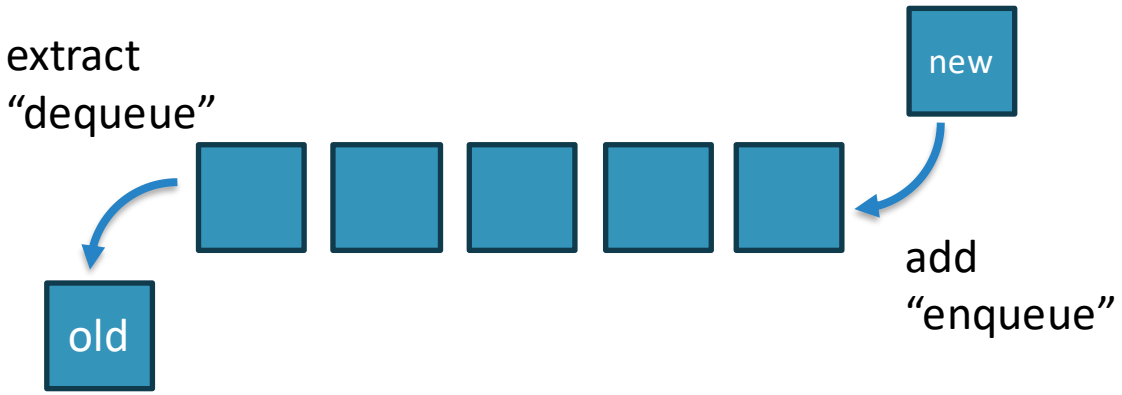
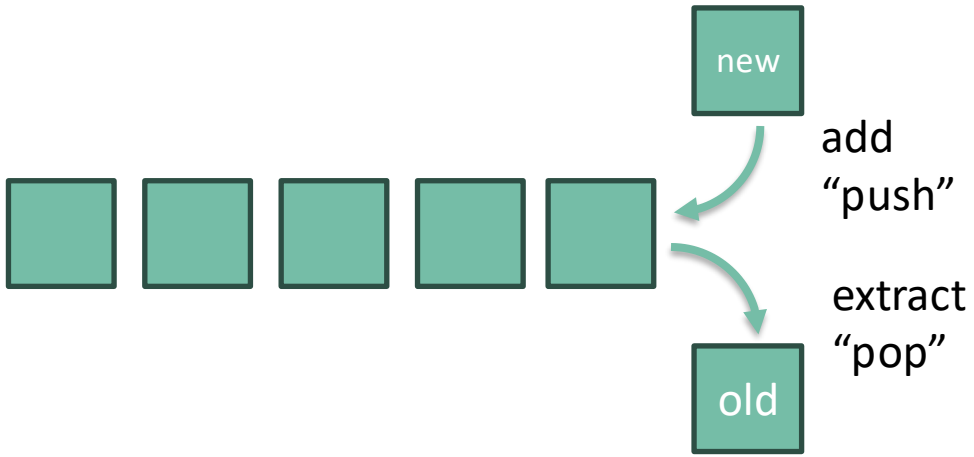
Queues and Stacks



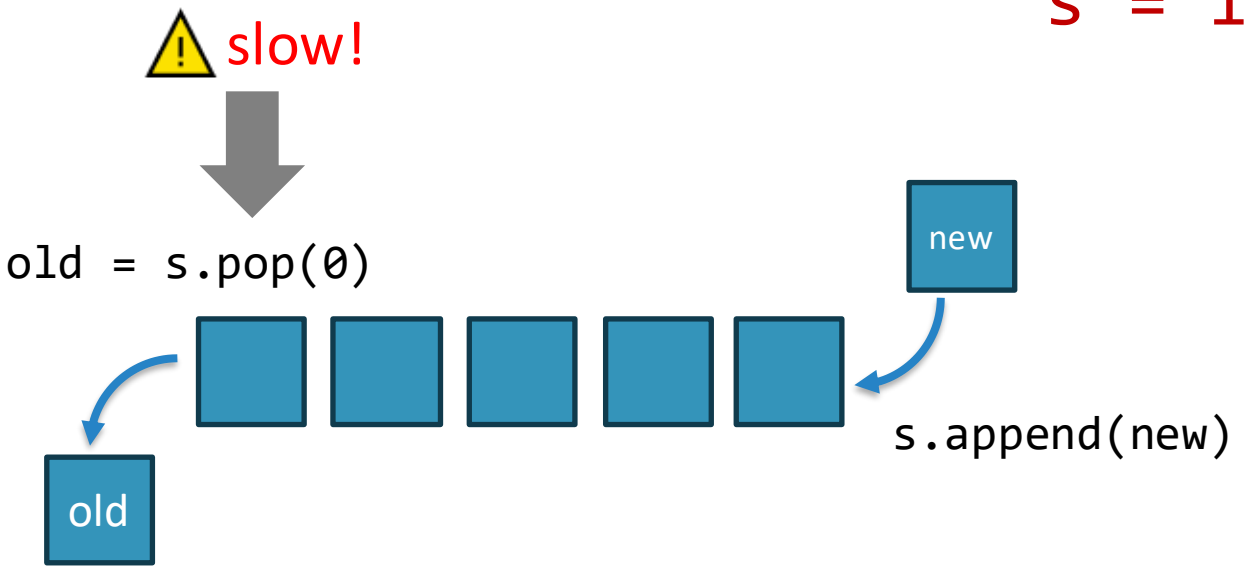
FIFO Queue – First-In First-Out



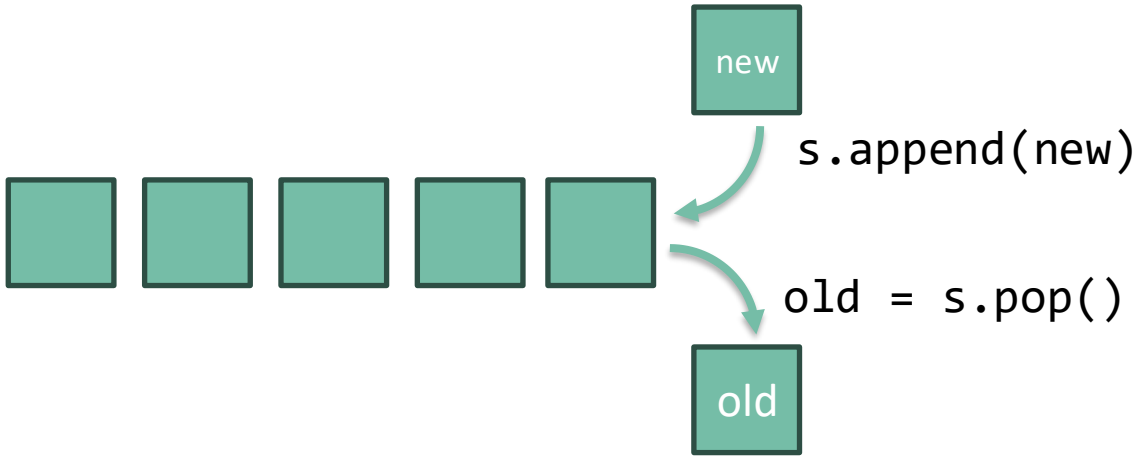
LIFO Stack – Larst-In First-Out

List implementations

```
s = list()
```



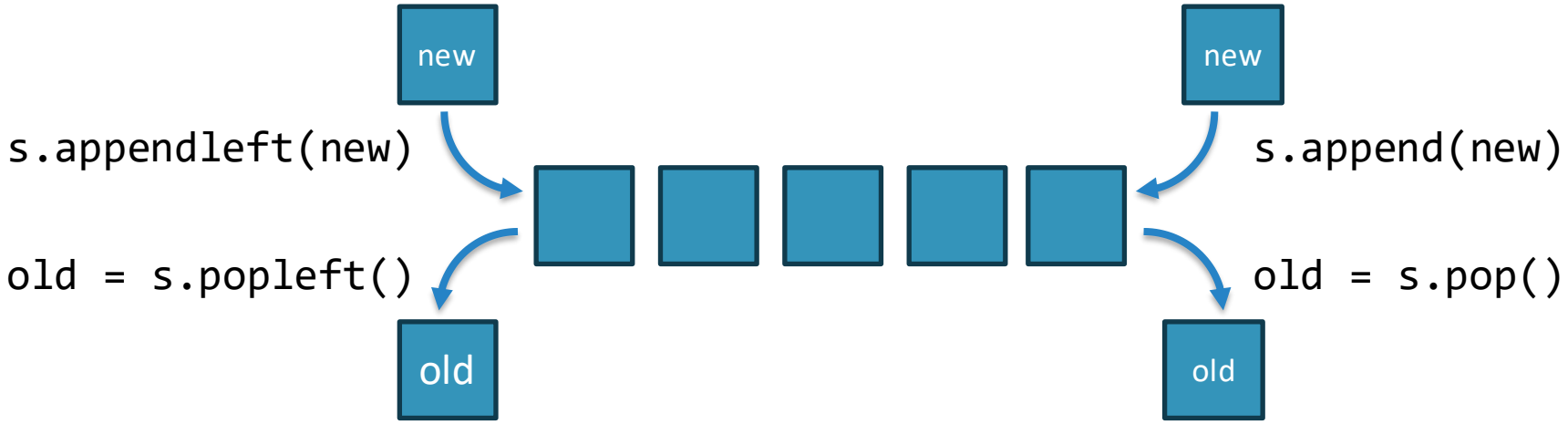
FIFO Queue – First-In First-Out



LIFO Stack – Last-In First-Out

deque: double-ended queue

```
s = collections.deque()
```



All operations have the same efficiency

https://docs.python.org/3/library/collections.html#deque-objects

Using a **deque**

As a FIFO Queue

- `append` and `popleft`
  - Most popular choice
- `appendleft` and `pop`
  - Also possible, same efficiency

As a LIFO Stack

- `append` and `pop`
  - Most popular choice
  - Might use a list, instead
- `appendleft` and `popleft`
  - Also possible, same efficiency

Other **deque** methods

`d = deque()`	New empty deque
`d = deque(iterable)`	Deque from list
`d = deque(maxlen=N)`	Hosts max N elements, discards older ones if more are added
`d.extend(iterable)`	Adds list of elements at end
`d.extendleft(iterable)`	Adds list of elements at beginning
`d.rotate(n)`	Rotate elements by n steps
`d[i]`	Access element (slower than lists)
`d.index(x), d.insert(i, x), d.remove(x), d.reverse()`	Same as lists

https://docs.python.org/3/library/collections.html#deque-objects

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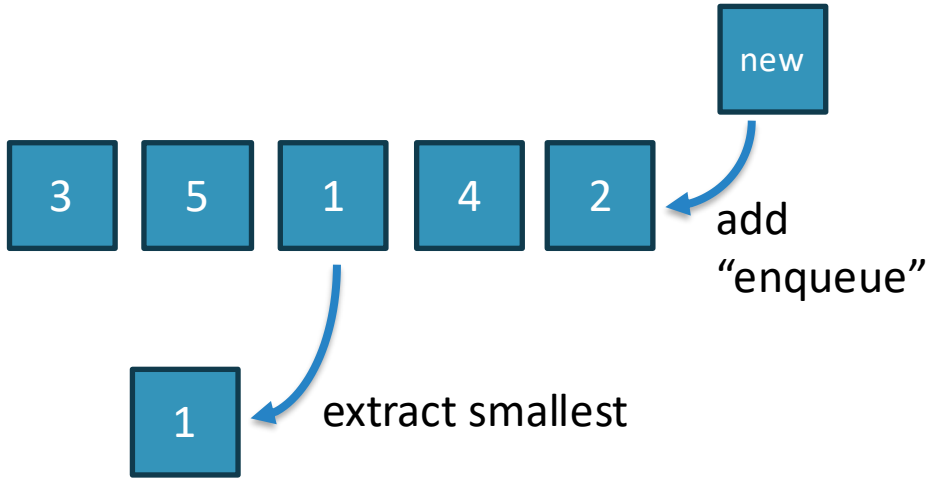
Priority Queues



- Elements are added in any order
- Elements are removed according to their "priority"
- Priority is determined by the sorting order of the elements
- Often, we create a tuple:
  - `(priority, value)`
- Or we rely on the object's `__lt__` method

Priority queues in Python

Items x must be comparable (implement __lt__)

heapq – uses plain lists

- `h = []`
- `h = heapify(iterable)`
- `len(h)`
- `len(h)==0`

- `heapq.heappush(h, x)`
- `x = heapq.heappop(h)`

queue.PriorityQueue

- `q = queue.PriorityQueue()`

- `q.qsize()`
- `q.empty()`
- `q.full()`
- `q.put(x)`
- `x = q.get_nowait()`

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